

İZMİR DEMOKRASİ UNIVERSITY



MIS203 – Basic Programming

Git and GitHub

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Git and GitHub

Git

A version control tool on your computer.

Records changes as commits.

Works without an internet connection.

GitHub

A website that hosts Git repositories.

Shares code and supports reviews.

Adds forks, pull requests and collaborators.

A GitHub account and a local Git installation serve different jobs.

From File to GitHub

A change moves through four places.

1. Working folder: edit a file.
2. Staging area: select changes with git add.
3. Local history: save a commit with git commit.
4. GitHub: publish commits with git push.

git pull moves remote commits back to the local repository.

Install Git

macOS

Check first: `git --version`

If missing: `xcode-select --install`

Terminal is enough for these commands.

Use the official Git install pages for current installers.

Windows

Install Git for Windows from git-scm.com/install/windows

Open Git Bash after installation.

Check: `git --version`

Identify Your Commits

Git needs a name and an email for local commits.

```
git --version  
git config --global user.name "Your Name"  
git config --global user.email "you@example.com"  
git config --global --list
```

How to update?

Windows: git update-git-for-windows

Apple: brew install git
brew upgrade git

Replace the sample name and email with your own GitHub account details.

Fork: Your Copy on GitHub

Use a fork when you need your own copy of another person's repository.

1. Open the instructor repository on GitHub.
2. Select Fork, then Create fork under your account.
3. Your fork has its own URL and its own main branch.

A fork does not give you write access to the instructor repository.

Clone Your Fork

A clone brings your fork from GitHub onto your computer.

```
git clone https://github.com/YOUR-USERNAME/MIS203-basic-programming.git  
cd MIS203-basic-programming  
git remote -v
```

Replace YOUR-USERNAME. The origin remote points to your fork.

Push to Your Fork

origin is your GitHub fork after cloning it.

```
git remote -v
```

```
git push origin main
```

Push sends your local main commits to your fork on GitHub.

Pull From Your Fork

Pull brings GitHub commits into your local branch.

```
git switch main  
git pull origin main  
git status
```

Here origin means your fork. Pull does not send local changes to GitHub.

Pull Request to the Instructor

The PR asks the instructor to review your changes.

1. In your fork, choose Contribute, then Open pull request.
2. Base: MehmethanErduran / main. Compare: your fork / main.
3. Title: Bug Fix - Your Name Surname. Explain all three fixes.
4. Check that only your folder changed, then create one PR.

Follow the live [bug-assignment/README.md](#) for the assignment rules.

A Branch Is a Separate Line of Work

In your own course repository, branch before changing a lab.

```
git switch main  
git pull origin main  
git switch -c week02-quote  
git branch --show-current
```

main stays stable while week02-quote holds the new work.

Push a Course Lab Branch

The lab file belongs in your own repository.

```
git add week02/lab-quiz/lab02_purchase_quote.py README.md  
git commit -m "Add Week 02 purchase quote"  
git push -u origin week02-quote
```

The first push sets the branch's upstream; later pushes can use git push.

Pull Request From Your Branch

A PR can also compare two branches in your own repository.

1. Open your own course repository on GitHub.
2. Select base: main and compare: week02-quote.
3. Read the file diff, describe the lab and request a review.
4. After merging, switch to main and pull its new commits.

This is a separate practice flow from the instructor's Bug Assignment PR.

Keep Your Fork Current

The instructor repository is the upstream remote.

```
git remote add upstream \  
https://github.com/MehmethanErduran/MIS203-basic-programming.git  
git fetch upstream  
git switch main  
git merge upstream/main  
git push origin main
```

Add upstream once. Fetch and merge when the instructor repository changes.

Collaborators Have Write Access

The repository owner invites another GitHub user.

1. Owner: Settings, Collaborators, Add people.
2. The invited person accepts the invitation.
3. A collaborator can push to the shared repository.
4. A fork works without collaborator access.

For this course, the Bug Assignment uses forks and pull requests.

Choose the Correct Workflow

Your Own Repository

Save weekly work under weekXX/.

Save labs under weekXX/lab-quiz/.

Push your branch; create a PR into your own main.

Instructor Repository

Fork it to your account.

Edit only your bug-assignment folder.

Push to your fork; send one PR to the instructor.

Collaborator access applies only after the owner sends an invitation.

When Git Reports a Problem

Check where you are and which remote you are using.

1. Permission denied: check git remote -v. Push to your fork.
2. Nothing to commit: check git status and your edits.
3. Push rejected: pull or sync before another push.

Read the full message before changing commands.

In-Class Practice

Use your own course repository for a safe first PR.

1. Create a branch for an existing weekXX/lab-quiz/ file.
2. Change one line, check git status and commit it.
3. Push the branch and open a PR into your own main.
4. Explain what the diff shows before merging.

Use the instructor repository only for the assigned Bug Assignment PR.

Official Links and Live Example

Git installation

git-scm.com/install/mac · git-scm.com/install/windows

GitHub guides

docs.github.com/en/get-started/using-git/about-git

docs.github.com/en/pull-requests/how-tos/work-with-forks

Course repository

<https://github.com/MehmethanErduran/MIS203-basic-programming>

See You Next Week